Radians

Chapter Overview

Solution 2

1. Angle measure

Convert between degrees and radians.

For example: express 45° in radians.

2. Length and area

Use a radius and an angle to find an arc length, sector perimeter or area.

3. Trigonometric equations

Solve $\sin x = \frac{1}{2}$ for $0 \leq x < \pi$.

4. Small angles

Explain why, for small θ in radians,

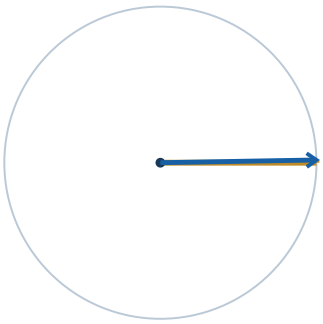
$\sin 5\theta + \tan 2\theta - \cos 2\theta \approx 2\theta^2 + 7\theta - 1$.

Radians

Solution 3

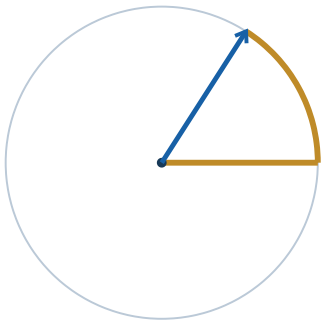
Degrees and radians

A degree is $1/360$ of a full turn. One radian is the angle whose arc length equals the radius.



Animation · 1°

1°: a very small turn.



Animation · 1 rad

1 rad ≈ 57.3°. Arc length = r.

A full circle has arc length $2\pi r$. How many radians is a complete turn?

Full turn

$360° = 2\pi$ rad

Why use radians as unit?

Solution 4

Why use radians?

Radians connect angle directly to length. They also give the simplest formulas in calculus.

$$\frac{d}{dx}(\sin x) = \cos x$$

This formula assumes that x is measured in radians.

A useful distinction

Degrees remain convenient for everyday geometry.

Radians are the standard angle measure in calculus and many mathematical models.

Always check the angle unit before using a trigonometric function.

Converting between radians and degrees

Solution 5

Degrees to radians

$\text{radians} = \text{degrees} \times \frac{\pi}{180}$.

Convert $90°, 45°, 135°, 72°$.

$90° = \pi/2$

$45° = \pi/4$

$135° = 3\pi/4$

$72° = 2\pi/5$

Radians to degrees

$\text{degrees} = \text{radians} \times \frac{180}{\pi}$.

Convert $\pi/3, \pi/6, 3\pi/2, 5\pi/6$.

$\pi/3 = 60°$

$\pi/6 = 30°$

$3\pi/2 = 270°$

$5\pi/6 = 150°$

Be able to convert common angles in your head...

Solution 6

Common angles

Give each angle in radians.

(a) $45°$

a $\pi/4$

(b) $60°$

b $\pi/3$

(c) $270°$

c $3\pi/2$

(d) $120°$

d
$2\pi/3$

(e) $30°$

e
$\pi/6$

(f) $135°$

f
$3\pi/4$

(g) $90°$

g
$\pi/2$

Anchor facts

$180° = \pi, \qquad 360° = 2\pi.$

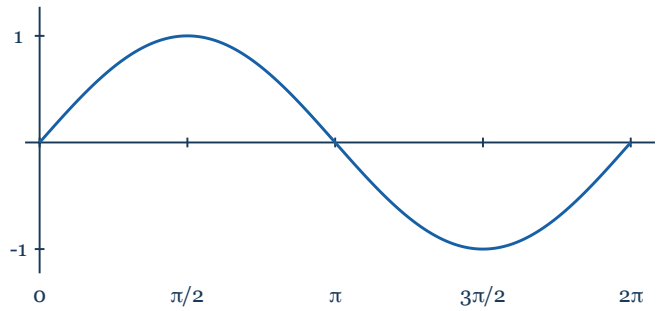
Graph Sketching with Radians

Solution 7

Sine

The curve keeps its shape; only the angle labels change.

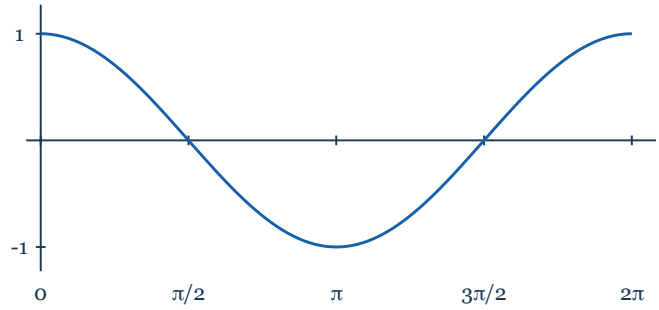
x	0	$\pi/2$	π	$3\pi/2$	2π
$\sin x$	0	1	0	-1	0



Cosine

One complete cycle spans 2π radians.

x	0	$\pi/2$	π	$3\pi/2$	2π
$\cos x$	1	0	-1	0	1



Test Your Understanding

Solution 8

Sketch $y = \cos(x + \pi/2)$ for $0 \le x < 2\pi$.

Start with $y = \cos x$ and translate it $\pi/2$ to the left.

Sketch

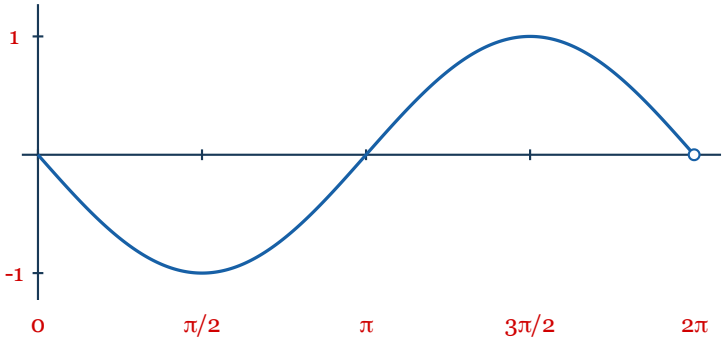
$\cos(x + \pi/2) = -\sin x$.

Key points:

$(0, 0),\ (\pi/2, -1),\ (\pi, 0),\ (3\pi/2, 1)$.

The curve approaches $(2\pi, 0)$, but this endpoint is excluded.

Sketch



sin, cos, tan of angles in radians

Solution 9

Symmetry and periodicity

$\sin x = \sin(\pi - x), \qquad \cos x = \cos(2\pi - x).$

Sine and cosine have period 2π; tangent has period π.

Find $\cos(4\pi/3)$ and $\sin(-7\pi/6)$.

$\cos(4\pi/3) = \cos 240^\circ = -\frac{1}{2}.$

2

$\sin(-7\pi/6) = \sin 150^\circ = \frac{1}{2}.$

Calculator mode

Set the angle unit to radians before evaluating trigonometric functions of radian angles.

As a quick check, $\sin(\pi/2)$ should give 1.

For exact common angles, use the known degree equivalents or unit-circle values.

Exercises 5A/5B – P2 Pages 116, 118

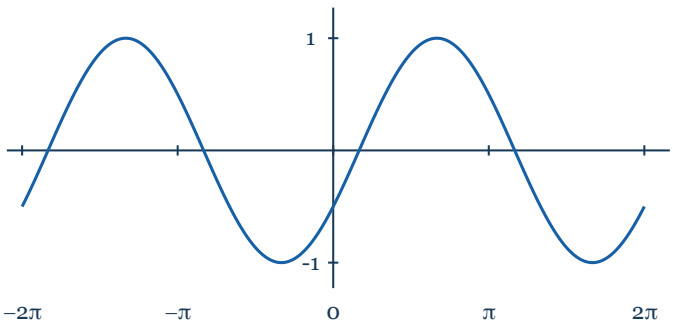
Solution 10

Exercise 5A, Question 8

The curve is

$$y = \cos\left(x - \frac{2\pi}{3}\right), \qquad -2\pi \le x \le 2\pi.$$

Find all intersections with the coordinate axes. (3 marks)



y-axis

For the y-axis, put $x = 0$:

$y = \cos(-2\pi/3) = -\frac{1}{2}.$

The intercept is $(0, -\frac{1}{2})$.

x-axis

For the x-axis:

$$x - \frac{2\pi}{3} = \frac{\pi}{2} + k\pi.$$

In the given interval the intercepts are

$$\left(-\frac{11\pi}{6}, 0\right), \left(-\frac{5\pi}{6}, 0\right),$$

$$\left(\frac{\pi}{6}, 0\right), \left(\frac{7\pi}{6}, 0\right).$$

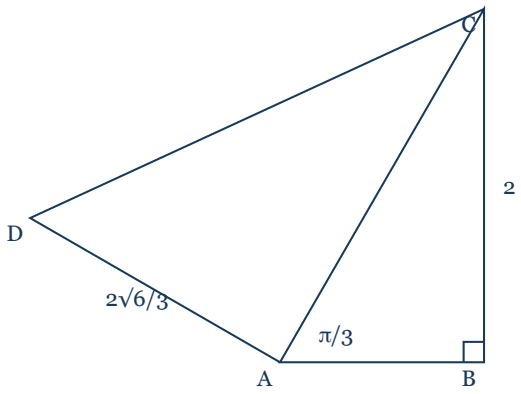
Exercises 5A/5B – P2 Pages 116, 118 — Solution 11

Exercise 5B, Question 3

Triangle ABC is right-angled at B, with $BC = 2$ and $\angle BAC = \pi/3$.

Triangle ACD is right-angled at A, with $AD = 2\sqrt{6}/3$.

Show that $DC = k\sqrt{2}$, and determine k.



Answer

In ABC:

$$AC = \frac{2}{\sin(\pi/3)} = \frac{4\sqrt{3}}{3}.$$

In ACD, by Pythagoras:

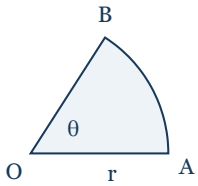
$$DC^2 = \left(\frac{2\sqrt{6}}{3}\right)^2 + \left(\frac{4\sqrt{3}}{3}\right)^2 = 8.$$

$$DC = 2\sqrt{2}, \qquad \boxed{k = 2}.$$

Arc length — Solution 12

Arc length

An angle of 1 radian subtends an arc of length r. An angle of θ radians therefore subtends what arc length?



Radians

$$\boxed{l = r\theta}$$

Use the same length units for l and r.

Compare degrees

An angle α in degrees is a fraction $\alpha/360$ of a full turn.

Degrees

$$l = \frac{\alpha}{360} \times 2\pi r.$$

Substituting $\alpha = 180\theta/\pi$ gives $l = r\theta$.

Examples

Solution 13

1. Find the arc length

A circle has radius 5.2 cm. An arc subtends 0.8 radians at the centre.

1

$l = r\theta = 5.2(0.8) = \boxed{4.16 \text{ cm}}$.

Calculator angle mode does not affect ordinary multiplication.

2. Find the angle

An arc AB of a circle with centre O and radius 7 cm has length 2.45 cm. Find $\angle AOB$.

2

$\theta = \dfrac{l}{r} = \dfrac{2.45}{7} = \boxed{0.35 \text{ rad}}$.

An angle "subtended" by an arc is the angle made by joining its endpoints to the centre.

Further Examples

Solution 14

1. Sector perimeter

A sector has radius r, angle θ radians and perimeter P. Express r in terms of P and θ.

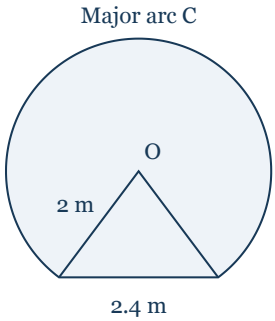
1

$P = 2r + r\theta = r(2 + \theta)$.

$\boxed{r = \dfrac{P}{2 + \theta}}$.

2. Garden pond

A pond is bounded by a straight chord $AB = 2.4$ m and the major arc of a circle of radius 2 m. Find the curved boundary length.



2

Bisect the minor central angle:

$x = \sin^{-1}(1.2/2) \approx 0.643501$.

Major angle: $\theta = 2\pi - 2x$.

$l = 2(2\pi - 2x) \approx \boxed{9.99 \text{ m}}$.

Test Your Understanding

Solution 15

Edexcel C2 January 2005, Q7

Triangle ABC has $AB = 8$ cm, $AC = 11$ cm, and $\angle BAC = 0.7$ radians.

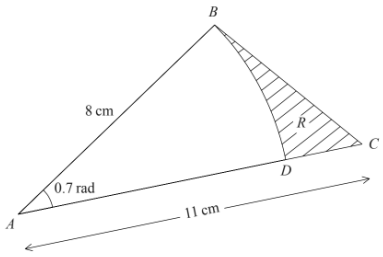
An arc BD, centre A and radius 8 cm, meets AC at D.

Region R is bounded by BC, CD and arc BD.

(a) Find arc BD.

(b) Find the perimeter of R, to 3 significant figures.

Figure 1



a

$BD = 8(0.7) = \boxed{5.6 \text{ cm}}$.

b

$CD = 11 - 8 = 3$ cm.

By the cosine rule:

$BC = \sqrt{8^2 + 11^2 - 2(8)(11)\cos 0.7}$.

$P = 5.6 + 3 + BC = \boxed{15.7 \text{ cm}}$ (3 s.f.).

Exercises 5C – Y2 Pages 120-122

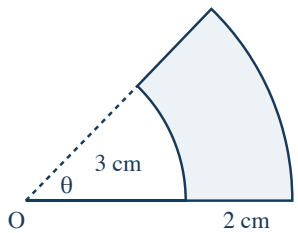
Solution 16

Exercise 5C, Question 5

An annular sector has inner radius 3 cm, outer radius 5 cm, and angle θ radians.

(a) Find its perimeter when $\theta = 0.8$.

(b) Find θ when its perimeter is 14 cm.



a

The two arcs have lengths 3θ and 5θ. Each straight edge is 2 cm.

$P = 3\theta + 5\theta + 2 + 2 = 8\theta + 4$.

At $\theta = 0.8$, $P = \boxed{10.4 \text{ cm}}$.

b

$8\theta + 4 = 14 \implies \theta = \dfrac{10}{8} = \boxed{1.25 \text{ rad}}$.

Exercise 5C, Question 6

A sector has radius r cm and angle 1.2 radians. It has the same perimeter as a square of area 36 cm^2.

Find r.

Answer

The square has side $\sqrt{36} = 6$ cm, so perimeter 24 cm.

The sector perimeter is

$2r + 1.2r = 3.2r$.

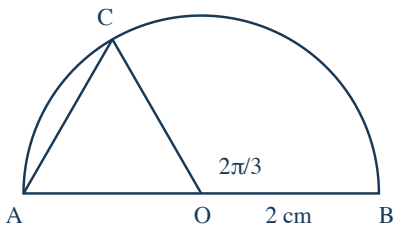
$3.2r = 24 \implies \boxed{r = 7.5 \text{ cm}}$.

Exercise 5C, Question 8

AB is the diameter of a circle of centre O, radius 2 cm. Point C is on the upper semicircle, with $\angle COB = 2\pi/3$.

(a) Find $\angle COA$. (1 mark)

(b) A brooch is bounded by chord AC, arc CB and diameter AB. Find its exact perimeter. (5 marks)



a

$\angle COA = \pi - \dfrac{2\pi}{3} = \boxed{\dfrac{\pi}{3}}$.

b

$AB = 4$. Arc $CB = 2(2\pi/3) = 4\pi/3$.

Triangle COA is equilateral, so $AC = 2$.

$\boxed{P = 6 + \dfrac{4\pi}{3} \text{ cm}}$.

Exercises 5C – Y2 Pages 120-122

Solution 19

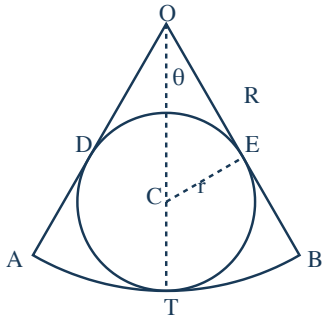
Exercise 5C, Question 10

Sector OAB has radius R and angle 2θ. A circle of centre C and radius r touches the arc at T and the two radii at D, E.

(a) Express OC in terms of R, r.

(b) Use right triangle OCE to show $R\sin\theta = r(1+\sin\theta)$.

(c) Given $\sin\theta = 3/4$ and sector perimeter 21 cm, find r to 3 significant figures.



a

$\boxed{OC = R - r}$.

b

$\sin\theta = \dfrac{r}{R-r}$.

$(R-r)\sin\theta = r \implies R\sin\theta = r(1+\sin\theta)$.

c

$r = 3R/7$, and $\theta = \sin^{-1}(3/4)$.

$2R + 2R\theta = 21 \implies R = \dfrac{21}{2(1+\theta)}$.

$\boxed{r = 2.43 \text{ cm}}$.

Sector Area

Solution 20

Sector area in degrees

A sector with angle α degrees is a fraction $\alpha/360$ of the circle.

Degrees

$A = \dfrac{\alpha}{360}\pi r^2$.

Sector area in radians

Substitute $\alpha = 180\theta/\pi$.

Radians

$A = \dfrac{180\theta/\pi}{360}\pi r^2 = \boxed{\dfrac{1}{2}r^2\theta}$.

The angle θ must be in radians.

Segment Area

Solution 21

Minor segment

A segment is bounded by a chord and an arc. Subtract the triangle between the radii from the sector.

For $0 < \theta < \pi$, use triangle area $\frac{1}{2}r^2 \sin\theta$.

Area

$$A_{\text{segment}} = A_{\text{sector}} - A_{\text{triangle}}$$
$$= \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin\theta$$
$$= \boxed{\frac{1}{2}r^2(\theta - \sin\theta)}.$$

Use radians throughout.

Examples

Solution 22

1. Find the radius

A minor sector has area 28.9 cm^2 and angle 0.8 radians.

1

$$28.9 = \frac{1}{2}r^2(0.8), \quad r^2 = 72.25.$$

$$\boxed{r = 8.5 \text{ cm}}.$$

2. Fenced plot

A sector-shaped plot has radius 55 m and total boundary length 176 m. Find its area.

2

$$\text{Arc length} = 176 - 2(55) = 66 \text{ m}.$$

$$\theta = \frac{66}{55} = 1.2.$$

$$A = \frac{1}{2}(55)^2(1.2) = \boxed{1815 \text{ m}^2}.$$

Segment Examples

Solution 23

1. Segment area

A minor segment has radius 4 m and chord length 5 m. Find its area.

1

$25 = 16 + 16 - 32\cos\theta, \quad \theta = \cos^{-1}(7/32).$

$A = 8(\theta - \sin\theta) = \boxed{3.00\ \text{m}^2}$ (3 s.f.).

2. Prove the relation

AB is a diameter of a circle of radius r. Point C lies on the upper semicircle, with $\angle BOC = \theta$.

The area of triangle AOC is three times the minor segment cut off by chord BC. Show $3\theta - 4\sin\theta = 0$.

2

$\angle AOC = \pi - \theta$, and $\sin(\pi - \theta) = \sin\theta$.

$\frac{1}{2}r^2\sin\theta = 3\left[\frac{1}{2}r^2(\theta - \sin\theta)\right]$.

$\sin\theta = 3\theta - 3\sin\theta \implies \boxed{3\theta - 4\sin\theta = 0}$.

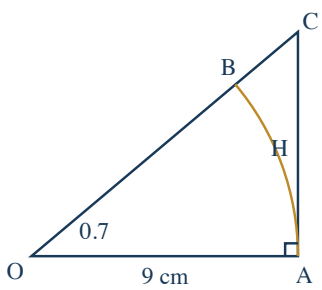
Test Your Understanding

Solution 24

Edexcel C2

Sector OAB has radius 9 cm and angle 0.7 radians. AC is perpendicular to OA, and O, B, C are collinear in that order.

Find (a) arc AB; (b) sector area; (c) length AC to 2 decimal places; (d) area H between arc AB and lines AC, CB, to 2 decimal places.



a $l = 9(0.7) = \boxed{6.3\ \text{cm}}$.

b $A = \frac{1}{2}(9^2)(0.7) = \boxed{28.35\ \text{cm}^2}$.

c $AC = 9\tan 0.7 = \boxed{7.58\ \text{cm}}$.

d $H = \frac{1}{2}(9)(9\tan 0.7) - 28.35 = \boxed{5.76\ \text{cm}^2}$.

Keep full precision until the final answer.

Exercise 5D, Question 6

A sector has radius 12 cm and area 100 cm^2. Find its perimeter.

Answer

$$100 = \frac{1}{2}(12)^2\theta \implies \theta = \frac{25}{18}.$$

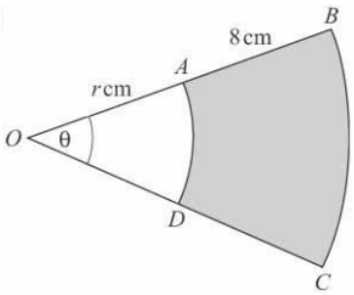
$$P = 24 + 12\left(\frac{25}{18}\right) = \boxed{\frac{122}{3}\text{ cm}}.$$

Exercise 5D, Question 12

Concentric arcs AD and BC, centre O, bound an annular sector. $OA = OD = r$ cm, $AB = DC = 8$ cm, and $\angle BOC = \theta$.

(a) Its area is 48 cm^2. Show $r = 6/\theta - 4$.

(b) Given $r = 10\theta$, find the perimeter.



a

$$\frac{1}{2}[(r+8)^2 - r^2]\theta = 48.$$

$$(8r + 32)\theta = 48 \implies (r+4)\theta = 6.$$

$$\boxed{r = \frac{6}{\theta} - 4}.$$

b

$10\theta^2 + 4\theta = 6$, so

$(5\theta - 3)(\theta + 1) = 0.$

Since $\theta > 0$, $\theta = 3/5$, $r = 6$.

$P = r\theta + (r+8)\theta + 16 = \boxed{28\text{ cm}}$.

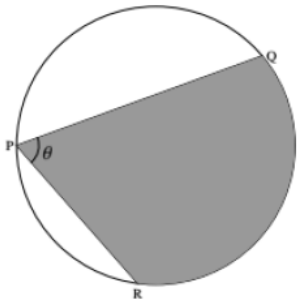
Extension

Solution 27

MAT 2012, Question 1J

Two chords PQ, PR of a unit circle meet at an angle θ at P.

Find the largest possible area bounded by the chords and the arc QR opposite P.



Position

The arc QR subtends 2θ at the centre. Its segment area is fixed.

Therefore maximise triangle PQR: its base is fixed, and its height is greatest when P lies on the perpendicular bisector of QR.

Area

In this symmetric arrangement, the region comprises a sector of angle 2θ and two triangles of area $\frac{1}{2}\sin\theta$.

$$A_{\max} = \frac{1}{2}(2\theta) + 2\left(\frac{1}{2}\sin\theta\right) = \boxed{\theta + \sin\theta}.$$

Solving Trigonometric Equations

Solution 28

1. Edexcel January 2007, Q6

Solve $2\cos^2 x + 1 = 5\sin x$ for $0 \le x < 2\pi$. Give exact answers.

1

$2(1 - \sin^2 x) + 1 = 5\sin x.$

$(2\sin x - 1)(\sin x + 3) = 0.$

Reject $\sin x = -3$. Thus

$$\boxed{x = \pi/6,\ 5\pi/6}.$$

2. Solve $\sin 3\theta = \sqrt{3}/2$ for $0 \le \theta \le 2\pi$.

2

First use $0 \le 3\theta \le 6\pi$.

$$3\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}, \frac{13\pi}{3}, \frac{14\pi}{3}.$$

Divide every value by 3:

$$\boxed{\theta = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{7\pi}{9}, \frac{8\pi}{9}, \frac{13\pi}{9}, \frac{14\pi}{9}}.$$

Exercise 5E, Question 9(a)

Solve $\cos(x - \pi/12) = 1/\sqrt{2}$ for $0 \le x < 2\pi$.

a

Let $u = x - \pi/12$, so $-\pi/12 \le u < 23\pi/12$.

$u = \pi/4,\ 7\pi/4$.

Add $\pi/12$: $\boxed{x = \pi/3,\ 11\pi/6}$.

9(b) Solve $\sin 3x = -1/2$ for $0 \le x < 2\pi$.

b

Use $0 \le 3x < 6\pi$:

$3x = \dfrac{7\pi}{6}, \dfrac{11\pi}{6}, \dfrac{19\pi}{6}, \dfrac{23\pi}{6}, \dfrac{31\pi}{6}, \dfrac{35\pi}{6}$.

Hence

$$\boxed{x = \frac{7\pi}{18}, \frac{11\pi}{18}, \frac{19\pi}{18}, \frac{23\pi}{18}, \frac{31\pi}{18}, \frac{35\pi}{18}}.$$

Exercise 5E, Question 10(a)

Solve $(1 + \tan\theta)(5\sin\theta - 2) = 0$ for $-\pi \le \theta < \pi$.

a

$\tan\theta = -1$ gives $-\pi/4,\ 3\pi/4$.

$\sin\theta = 2/5$ gives $\sin^{-1}(2/5)$ and $\pi - \sin^{-1}(2/5)$.

Thus $\boxed{\theta = -\pi/4,\ 0.412,\ 3\pi/4,\ 2.73}$, with decimals to 3 significant figures.

10(b) Solve $4\tan x = 5\sin x$ for $0 \le x < 2\pi$.

b

Where $\cos x \neq 0$,

$\sin x(4 - 5\cos x) = 0$.

Either $\sin x = 0$, giving $0, \pi$, or $\cos x = 4/5$, giving $0.644, 5.64$.

All four satisfy the original equation.

Exercise 5E, Question 13

Show that $8\sin^2 x + 4\sin x - 20 = 4$ has no real solutions.

Answer

$8\sin^2 x + 4\sin x - 24 = 0.$

$(2\sin x - 3)(\sin x + 2) = 0.$

This requires $\sin x = 3/2$ or $\sin x = -2.$

Both lie outside $[-1, 1]$, so there are no real solutions.

Exercise 5E, Question 15 — measurements as printed

In triangle ABC, $AB = 5$ cm, $AC = 12$ cm, $\angle ABC = 0.5$ radians, and $\angle ACB = x$.

(a) Find $\sin x$, to 3 decimal places.

(b) Find the possible triangle angle x, to 2 decimal places.

a

By the sine rule:

$\frac{\sin x}{5} = \frac{\sin 0.5}{12}.$

$\boxed{\sin x = 0.200}$ (3 d.p.).

b

$x_1 = \sin^{-1}(5\sin 0.5/12) \approx 0.2011.$

The second sine solution is $\pi - x_1$, but it exceeds $\pi - 0.5$, so it cannot be a triangle angle.

$\boxed{x = 0.20 \text{ rad}}.$

With these measurements there is only one triangle.

Extension

Solution 33

MAT 2010, Question 1C

How many solutions in $0 \leq x \leq 2\pi$ satisfy

$\sin^2 x + 3\sin x \cos x + 2\cos^2 x = 0$?

Answer

$(\sin x + 2\cos x)(\sin x + \cos x) = 0.$

$\cos x = 0$ is not a solution. Divide by $\cos x$:

$\tan x = -2 \quad$ or $\quad \tan x = -1.$

Each has two solutions in the interval. They are distinct, so the total is $\boxed{4}$.

Small Angle Approximations

Solution 34

When the angle is small

For θ near zero, measured in radians:

$$\boxed{\sin\theta \approx \theta, \quad \tan\theta \approx \theta}$$

$$\boxed{\cos\theta \approx 1 - \frac{\theta^2}{2}}.$$

These approximations improve as θ approaches zero.

The unit matters

Near zero, the graph $y = \sin x$ is close to $y = x$ only when x is measured in radians.

For example, $\sin(0.1) \approx 0.0998334$, whereas $\sin(0.1^\circ) \approx 0.00174533$.

Small Angle Approximations

Solution 35

Geometric explanation

For a unit circle and $0 < \theta < \pi/2$:

Inner triangle area $= \frac{1}{2}\sin\theta$.

Sector area $= \frac{1}{2}\theta$.

Outer tangent triangle area $= \frac{1}{2}\tan\theta$.

Comparing areas gives

$\sin\theta < \theta < \tan\theta$.

Reasoning

Dividing by $\theta > 0$ gives

$$\cos\theta < \frac{\sin\theta}{\theta} < 1.$$

Both bounds approach 1 as $\theta \to 0$. Hence

$$\frac{\sin\theta}{\theta} \to 1, \qquad \sin\theta \approx \theta.$$

Radians are essential because the sector area used was $\theta/2$.

Examples

Solution 36

1. Small non-zero θ, in radians

Approximate $\dfrac{\sin 2\theta + \tan\theta}{2\theta}$.

1

$$\frac{2\theta + \theta}{2\theta} = \boxed{\frac{3}{2}}.$$

2. Approximate $\dfrac{\cos 4\theta - 1}{\theta \sin 2\theta}$.

2

$$\frac{1 - (4\theta)^2/2 - 1}{2\theta^2} = \boxed{-4}.$$

3. Show $\sin 5\theta + \tan 2\theta - \cos 2\theta \approx 2\theta^2 + 7\theta - 1$, then state its limiting value as $\theta \to 0$.

3

$$5\theta + 2\theta - \left(1 - \frac{(2\theta)^2}{2}\right) = 2\theta^2 + 7\theta - 1.$$

As $\theta \to 0$, this approaches $\boxed{-1}$.

Exercise 5F, Question 2(a)

For small non-zero θ in radians, show $\dfrac{\sin 3\theta}{\theta \sin 4\theta} \approx \dfrac{3}{4\theta}$.

a

$$\frac{\sin 3\theta}{\theta \sin 4\theta} \approx \frac{3\theta}{\theta(4\theta)} = \boxed{\frac{3}{4\theta}}.$$

2(b) Show $\dfrac{\cos\theta - 1}{\tan 2\theta} \approx -\dfrac{\theta}{4}$.

b

$$\frac{1 - \theta^2/2 - 1}{2\theta} = \frac{-\theta^2/2}{2\theta} = \boxed{-\frac{\theta}{4}}.$$

Exercise 5F, Question 4

The approximation $\sin\theta \approx \theta$ has percentage error 1%. For small positive θ, show $100\theta = 101\sin\theta$.

4

$$\frac{\theta - \sin\theta}{\sin\theta} \times 100 = 1.$$

$$100\theta - 100\sin\theta = \sin\theta.$$

$$\boxed{100\theta = 101\sin\theta}.$$

Question 5

For small θ, simplify $\dfrac{4\cos 3\theta - 2 + 5\sin\theta}{1 - \sin 2\theta}$.

Hence state its limiting value.

5

$$\frac{4(1 - 9\theta^2/2) - 2 + 5\theta}{1 - 2\theta} = \frac{2 + 5\theta - 18\theta^2}{1 - 2\theta}.$$

$$= \frac{(1 - 2\theta)(2 + 9\theta)}{1 - 2\theta} = 2 + 9\theta.$$

The limiting value is $\boxed{2}$.

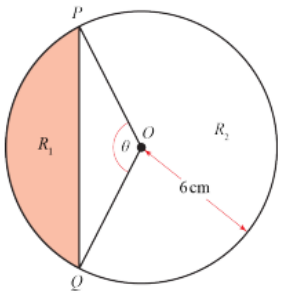
Mixed Exercises – Y2 Page 135 — Solution 39

Mixed Exercise, Question 7

A chord PQ in a circle of radius 6 cm subtends a minor angle θ at centre O. It divides the circle into minor segment R_1, area A_1, and major segment R_2, area A_2.

(a) Show $A_1 = 18(\theta - \sin\theta)$.

(b) If $A_2 = 3A_1$, show $\sin\theta = \theta - \pi/2$.



a

$$A_1 = \frac{1}{2}(6)^2\theta - \frac{1}{2}(6)^2\sin\theta = \boxed{18(\theta - \sin\theta)}.$$

b

The circle area is $36\pi = A_1 + A_2 = 4A_1$.

$36\pi = 72(\theta - \sin\theta)$.

$$\boxed{\sin\theta = \theta - \frac{\pi}{2}}.$$

Mixed Exercises– Y2 Page 135 — Solution 40

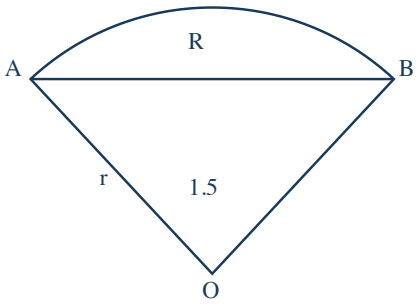
Mixed Exercise, Question 9

A sector OAB has radius r cm, angle 1.5 radians and area 15 cm^2.

(a) Prove $r = 2\sqrt{5}$.

(b) Find the sector perimeter.

(c) Find the area of the segment between arc AB and chord AB, to 3 decimal places.



a

$$15 = \frac{1}{2}r^2(1.5) \implies r^2 = 20 \implies \boxed{r = 2\sqrt{5}}.$$

b

$$P = 2r + 1.5r = 3.5(2\sqrt{5}) = \boxed{7\sqrt{5}\text{ cm}}.$$

c

$$A = 15 - \frac{1}{2}(20)\sin 1.5 = \boxed{5.025\text{ cm}^2}.$$